

For given

$$\mathcal{X} = \{0, 1, \dots, N-1\} = \mathcal{X}_{\text{good}} \cup \mathcal{X}_{\text{bad}}, \text{ where } \mathcal{X}_{\text{good}} \cap \mathcal{X}_{\text{bad}} = \emptyset$$

$$A|0\rangle = \sum_{j \in \mathcal{X}} \alpha_j |j\rangle = \sin \theta |\psi_{\text{good}}\rangle + \cos \theta |\psi_{\text{bad}}\rangle =: |\psi\rangle, \text{ where } (\sin \theta)^2 = \sum_{j \in \mathcal{X}_{\text{good}}} |\alpha_j|^2 \text{ and } (\cos \theta)^2 = \sum_{j \in \mathcal{X}_{\text{bad}}} |\alpha_j|^2$$

$$V|\psi_{\text{good}}\rangle = -|\psi_{\text{good}}\rangle \text{ and } V|\psi_{\text{bad}}\rangle = |\psi_{\text{bad}}\rangle$$

The task is to estimate the amplitude $\sin \theta \in [0, 1]$. Amplitude “ $\sin \theta$ ” estimation $\leftrightarrow$ phase “ θ ” estimation

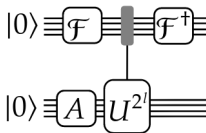
$$\mathcal{U} = W V, \text{ where } W = 2|\psi\rangle\langle\psi| - I,$$

$$\mathcal{U} \begin{pmatrix} |\psi_{\text{good}}\rangle \\ |\psi_{\text{bad}}\rangle \end{pmatrix} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} \begin{pmatrix} |\psi_{\text{good}}\rangle \\ |\psi_{\text{bad}}\rangle \end{pmatrix}$$

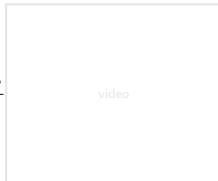
$$\mathcal{U} |\psi_{\pm}\rangle = e^{\pm i 2\theta} |\psi_{\pm}\rangle$$

$$|\psi_{\pm}\rangle = \frac{\mp i |\psi_{\text{good}}\rangle + |\psi_{\text{bad}}\rangle}{\sqrt{2}}$$

$$|\psi\rangle = \frac{e^{i\theta} |\psi_{+}\rangle + e^{-i\theta} |\psi_{-}\rangle}{\sqrt{2}}$$



$$|0\rangle|\psi\rangle \rightarrow \frac{e^{i\theta} |\frac{\theta}{\pi}\rangle |\psi_{+}\rangle + e^{-i\theta} |-\frac{\theta}{\pi}\rangle |\psi_{-}\rangle}{\sqrt{2}}$$



P. Kaye, R. Laflamme, and Michele Mosca, An Introduction to Quantum Computing, Chapt 8.

P. Rebentrost, B. Gupt, and T. R. Bromley, Phys. Rev. A **98**, 022321 (2018).

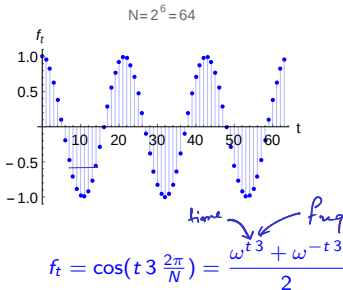
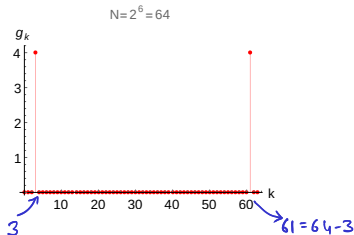
Discrete Fourier transform (DFT)

For n qubits we have $N = 2^n$ and $\omega = \exp(i \frac{2\pi}{N})$ is N th root of unity.

$$\underbrace{\begin{pmatrix} g_0 \\ g_1 \\ \vdots \\ g_{N-1} \end{pmatrix}}_{\mathbf{g}} = \frac{1}{\sqrt{N}} \underbrace{\begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega & \cdots & \omega^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \cdots & \omega^{(N-1)(N-1)} \end{pmatrix}}_{\mathbf{F}_N} \underbrace{\begin{pmatrix} f_0 \\ f_1 \\ \vdots \\ f_{N-1} \end{pmatrix}}_{\mathbf{f}} \quad \text{takes } O(N^2) = O(2^{2n}) \text{ steps and finds period.}$$

$N \times N$

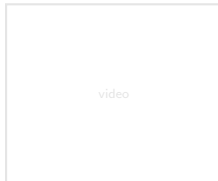
$t = 0, 1, 2, \dots, N-1 = 63$



$$\text{Time Period} = \frac{1}{f} = \frac{1}{3}$$

Uses: real-world data processing

QFT $\sim O(n^2)$



Fast Fourier transform (FFT)

$$t = 257 = 200 + 50 + 7 = 2 \times 10^2 + 5 \times 10^1 + 7 \times 10^0$$

$$= \underbrace{2}_{256} + 1 = 1 \times 2^8 + 0 \times 2^7 + \dots + 0 \times 2^1 + \underbrace{1}_{1} \times 2^0$$

$$= 100000001$$

For n qubits we have $N = 2^n$ and $\omega = \exp(i \frac{2\pi}{N})$ is N th root of unity.

Binary (n -bit) representation: $t = t_1 2^{n-1} + t_2 2^{n-2} + \dots + t_n 2^0 \equiv \underline{t_1 t_2 \dots t_n}$ for $t < 2^n = 9$ $t_i \in \{0, 1\}$

$$k = 0, 1, \dots, N-1$$

DFT: $g_k = \frac{1}{\sqrt{N}} \sum_{t=0}^{N-1} \omega^{kt} f_t$ one summation, which is adding $N = 2^n$ terms $\hookrightarrow O(N^2)$ steps

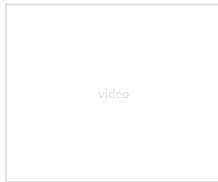
$\omega^{kt_1 2^{n-1}} \omega^{kt_2 2^{n-2}} \dots \omega^{kt_n}$

FFT: $g_k = \frac{1}{\sqrt{N}} \sum_{t_1=0}^1 \dots \sum_{t_n=0}^1 \omega^{k(t_1 2^{n-1} + t_2 2^{n-2} + \dots + t_n 2^0)} f_{t_1 t_2 \dots t_n}$ n summations, each one is adding two terms

$$= \frac{1}{\sqrt{N}} \sum_{t_1=0}^1 \omega^{k t_1 2^{n-1}} \dots \sum_{t_n=0}^1 \omega^{k t_n 2^0} f_{t_1 t_2 \dots t_n}$$
 divide and conquer

FFT takes $O(\underbrace{n 2^n}_N) = O(\underbrace{\log(N) N}_N)$ steps.

J. W. Cooley and J. W. Tukey, Math. Comput. 19, 297 (1965).



Quantum Fourier transform (QFT)

$k_i = 0, 1$

For n qubits we have $N = 2^n$ and $\omega = \exp(i \frac{2\pi}{N})$ is N th root of unity.

Binary (n -qubit) representation: $k = k_1 2^{n-1} + k_2 2^{n-2} + \dots + k_n 2^0 \Rightarrow |k\rangle \equiv |k_1 k_2 \dots k_n\rangle$

$$F^\dagger F = I = F F^\dagger$$

$$\begin{aligned} \text{QFT: } \mathcal{F}|t\rangle &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{tk} |k\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{k_1=0}^1 \dots \sum_{k_n=0}^1 \omega^{t(k_1 2^{n-1} + k_2 2^{n-2} + \dots + k_n 2^0)} |k_1 k_2 \dots k_n\rangle \\ &= \frac{1}{\sqrt{N}} \left[\sum_{k_1=0}^1 \omega^{t k_1 2^{n-1}} |k_1\rangle \right] \otimes \dots \otimes \left[\sum_{k_n=0}^1 \omega^{t k_n 2^0} |k_n\rangle \right] \end{aligned}$$

$|k_1\rangle \otimes |k_2\rangle \otimes \dots \otimes |k_n\rangle$

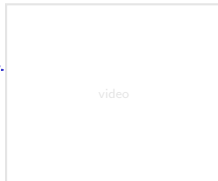
Power of superposition
(a.m)
(a.c)

$$\begin{aligned} \sum_{k=0}^{N-1} g_k |k\rangle &= \mathcal{F} \left[\sum_{t=0}^{N-1} f_t |t\rangle \right] \\ &= \sum_{t=0}^{N-1} \underbrace{\left[\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{tk} f_t |k\rangle \right]}_{g_k} \end{aligned}$$

Exponential data compression: storing $N = 2^n$ complex numbers $\{f_t\}$ in n qubits.

Exponentially fast: ~~QFT~~ takes $O(n^2) = O(\log(N)^2)$ computation steps.

Using Superposition.
 $|14\rangle$



Quantum Fourier transform (QFT)

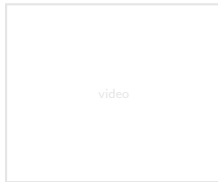
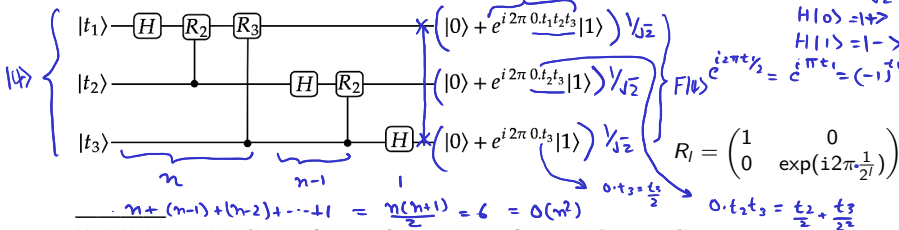
$$N = 2^n, \quad \omega^{t 2^{(n-l)}} = \exp(i \underbrace{\frac{2\pi}{N} t 2^{(n-l)}}_{2\pi \frac{t}{2^{n-(n-l)}}}) = \exp(i 2\pi \underbrace{\frac{t}{2^l}}_{\text{Integer}}) \text{ and } \frac{t}{2^l} = \underbrace{t_1 2^{n-1-l} + \dots + t_{n-l} 2^0}_{\text{Integer}} + \underbrace{t_{n-l+1} \frac{1}{2} + \dots + t_n \frac{1}{2^l}}_{\text{last } l \text{ bits of } t}$$

$$\mathcal{F}|t\rangle = \frac{1}{\sqrt{N}} \left[\sum_{k_1=0}^1 \omega^{t k_1 2^{n-1}} |k_1\rangle \right] \otimes \dots \otimes \left[\sum_{k_n=0}^1 \omega^{t k_n 2^0} |k_n\rangle \right] = \frac{1}{\sqrt{2}} [|0\rangle + \omega^{t 2^{n-1}} |1\rangle] \otimes \dots \otimes \frac{1}{\sqrt{2}} [|0\rangle + \omega^{t 2^0} |1\rangle]$$

$$\mathcal{F}|t_1 t_2 \dots t_n\rangle = \frac{1}{\sqrt{2}} [|0\rangle + e^{i 2\pi 0.t_n} |1\rangle] \otimes \dots \otimes \frac{1}{\sqrt{2}} [|0\rangle + e^{i 2\pi 0.t_1 \dots t_n} |1\rangle]$$

$$0.257 = 0.2 + 0.05 + 0.007 \\ = \frac{2}{10} + \frac{5}{10^2} + \frac{7}{10^3}$$

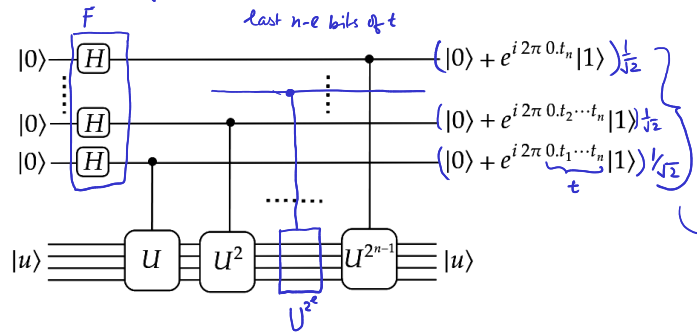
$n=3$



Q π_{AI} $t \in \{0,1\}$ Unknown phase (eigenvalue) estimation

$$t \approx 0.t_1 t_2 \dots t_n = \frac{t_1}{2} + \frac{t_2}{2^2} + \dots + \frac{t_n}{2^n} \in [0, 1)$$

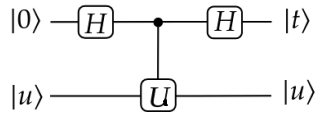
$$t 2^l \approx \underbrace{t_1 2^{l-1} + \dots + t_l}_{\text{Integer}} + \underbrace{\frac{t_{l+1}}{2} + \dots + \frac{t_n}{2^{n-l}}}_{0.t_{l+1} \dots t_n}$$



$$U^{2^e}|u\rangle = e^{i2\pi t \frac{2^e}{2^n}}|u\rangle$$

Phase kickback: $e^{i\theta}$ $\theta \in \mathbb{R}$
 $i = \sqrt{-1}$

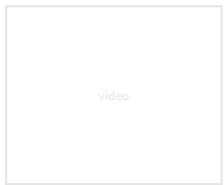
$$U|u\rangle = e^{i2\pi t}|u\rangle$$



$$|0\rangle|u\rangle \rightarrow \frac{1}{\sqrt{2}}[|0\rangle|u\rangle + |1\rangle|u\rangle] \rightarrow \frac{1}{\sqrt{2}}[|0\rangle + e^{i2\pi t}|1\rangle]|u\rangle$$

$$|0\rangle|u\rangle \mapsto \left(\frac{1+e^{i2\pi t}}{2}|0\rangle + \frac{1-e^{i2\pi t}}{2}|1\rangle\right)|u\rangle$$

$$F^{-1} \rightarrow |t_1 t_2 \dots t_n\rangle$$



Uses: Shor factoring algorithm and Quantum HHL algorithm for system of linear eqs.